Economics S5040: Dynamic Stochastic Modeling
Problem Set 3 — Matrices, Monte Carlo Methods, and Markov Chains

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Instructions

Submit one Jupyter notebook containing all derivations, Python code, figures, numerical answers, and economic explanations.

You may consult lecture notes, classmates, textbooks, and AI tools. However, your notebook must make clear that you understand every result you report. A numerical answer without an explanation will receive little credit.

For computational questions, do not simply print an answer. Explain what the calculation means economically.

Problem 1. Matrix Multiplication and Economic Accounting

Suppose an economy has three sectors: agriculture, manufacturing, and services. Current output is represented by

$$x = \begin{pmatrix} 100 \\ 200 \\ 300 \end{pmatrix}.$$

Suppose

$$A = \begin{pmatrix} 0.10 & 0.20 & 0.00 \\ 0.30 & 0.10 & 0.20 \\ 0.00 & 0.20 & 0.40 \end{pmatrix}.$$

(a) State the dimensions of A and x. Is Ax defined? Is xA defined if x is treated as a column vector? Explain.

(b) Compute Ax by hand, showing explicitly the row-by-column calculation for each entry. Then verify your answer in Python using `@`.

(c) Compute `A * x` in Python. Explain why this is not matrix multiplication and why its result has a different interpretation.

(d) Let

$$B = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

Without asking Python first, determine whether BA and AB are defined and state the dimensions of any product that is defined. Then verify in Python.

(e) In a short paragraph, explain the row-by-column rule in words. Your explanation should be clear enough for a student who has never taken linear algebra.

Problem 2. Monte Carlo Integration

Let X be uniformly distributed on $[0, 1]$ and define

$$g(X) = e^{X^2}.$$

We want to calculate

$$\mu = E[g(X)] = \int_0^1 e^{x^2}\, dx.$$

This integral does not have an elementary antiderivative.

(a) Use a numerical integration routine in Python to obtain an accurate benchmark value for μ.

(b) Set the random seed to 5040. Draw N independent observations from the uniform distribution and compute

$$\widehat{\mu}_N = \frac{1}{N} \sum_{i=1}^{N} e^{X_i^2}$$

for

$$N = 10,\ 100,\ 1{,}000,\ 10{,}000,\ 100{,}000.$$

Create a table containing N, $\widehat{\mu}_N$, and the absolute error $|\widehat{\mu}_N - \mu|$.

(c) Plot the Monte Carlo estimate against N. Add a horizontal line at the benchmark value μ. Use a logarithmic scale for the horizontal axis.

(d) The error need not fall every time N increases. Explain why this does not contradict the Law of Large Numbers.

(e) Now repeat the experiment 1,000 times for each of

$$N = 100, \quad N = 1{,}000, \quad N = 10{,}000.$$

For each N, compute the standard deviation across the 1,000 Monte Carlo estimates. What happens when N is multiplied by 100? Relate your numerical finding to the approximate $1/\sqrt{N}$ behavior of Monte Carlo sampling error.

Problem 3. Employment as a Markov Chain

Suppose an individual's labor-market state is either employed (E) or unemployed (U), with transition matrix

$$P = \begin{pmatrix} 0.90 & 0.10 \\ 0.40 & 0.60 \end{pmatrix}.$$

Rows represent today's state and columns tomorrow's state.

(a) Verify that P is a stochastic matrix. Explain what each of its four entries means.

(b) Compute P^2 by hand and verify it in Python. Explain in words why

$$(P^2)_{EU} = P_{EE}P_{EU} + P_{EU}P_{UU}.$$

(c) Compute P^{10}. If a person is employed today, what is the probability that the person is unemployed ten periods from now? What if the person is unemployed today?

(d) Starting from

$$\psi_0 = (1, 0),$$

calculate $\psi_t = \psi_0 P^t$ for $t = 0, \ldots, 30$. Plot the unemployment probability against t.

Repeat the exercise starting from

$$\widetilde{\psi}_0 = (0, 1).$$

Put both paths on the same graph.

(e) Solve analytically for the stationary distribution

$$\psi^* = \psi^* P.$$

Verify your answer numerically.

(f) Explain why the two paths in part (d) approach the same value. Be careful: convergence of the population distribution does not mean that an individual worker eventually stops changing labor-market states.

Problem 4. Simulation Versus the Stationary Distribution

Continue with the employment transition matrix from Problem 3.

(a) Write your own Python function that simulates one individual's employment history for T periods. Do not use a QuantEcon Markov-chain simulation routine. Your function should use the appropriate row of P to draw the next state.

(b) Set the seed to 5040, start the worker employed, and simulate $T = 100{,}000$ periods. Calculate the fraction of periods spent employed and unemployed. Compare these frequencies with the stationary distribution.

(c) Repeat the experiment for

$$T = 100, \quad 1{,}000, \quad 10{,}000, \quad 100{,}000.$$

Plot the simulated unemployment frequency against T, together with a horizontal line showing the stationary unemployment rate.

(d) Repeat the $T = 1{,}000$ experiment 500 times. Plot a histogram of the 500 simulated unemployment frequencies and mark the stationary unemployment rate.

(e) Explain the connection between this exercise and Monte Carlo methods. What is random in the simulation? What is the theoretical object that the simulation is approximating?

Problem 5. A Three-State Business-Cycle Model

Aggregate productivity can be low (L), normal (N), or high (H). Its transition matrix is

$$P = \begin{pmatrix} 0.70 & 0.25 & 0.05 \\ 0.10 & 0.80 & 0.10 \\ 0.05 & 0.25 & 0.70 \end{pmatrix}.$$

Output in the three states is

$$y = \begin{pmatrix} 80 \\ 100 \\ 130 \end{pmatrix}.$$

(a) Explain the economic meaning of P_{13}, P_{22}, and P_{31}.

(b) Suppose the economy is in the low state today. Compute the probability distribution over states one period, two periods, and ten periods from now.

(c) Find the stationary distribution numerically. Verify that its entries sum to one and that

$$\psi^* P = \psi^*.$$

(d) Calculate long-run mean output

$$\bar{y} = \psi^* y.$$

Explain why this is a matrix product and why it is also an expectation.

(e) Starting from each of the three possible initial states, compute expected output t periods ahead for $t = 0, \ldots, 25$. Plot all three paths on one figure. What common value do they approach?

(f) Now change only the persistence of the extreme states to

$$\widetilde{P} = \begin{pmatrix} 0.90 & 0.05 & 0.05 \\ 0.10 & 0.80 & 0.10 \\ 0.05 & 0.05 & 0.90 \end{pmatrix}.$$

Find its stationary distribution and long-run mean output. Compare the result with the original model.

Is greater persistence by itself enough to tell you whether long-run mean output must rise or fall? Explain.

Problem 6. Synthesis: Distribution Calculations and Simulation

This problem asks you to compare two ways of studying the three-state model in Problem 5.

(a) Starting from the low state, use matrix multiplication to compute the exact probability of being in each state after 20 periods.

(b) Starting from the low state, simulate 100,000 independent 20-period histories. Estimate the probability of each terminal state from the simulated sample.

(c) Compare the simulated probabilities with the exact probabilities. Report the error for each state.

(d) Increase the number of simulations from 100 to 100,000 and produce a figure showing how the Monte Carlo estimate of the probability of being in the high state after 20 periods approaches the exact value.

(e) Explain the conceptual difference between computing

$$\psi_0 P^{20}$$

and estimating the same object by simulation. Why might an economist use simulation when matrix multiplication gives an exact answer in this example?

A Final Requirement

At the end of your notebook include a short section entitled

What surprised me?

Identify one result from this assignment that you did not find obvious before doing the calculations. Explain why it surprised you and what you now understand differently.